Detecting Public Perceptions of Apollo Go on Chinese Social Media Based on Topic Modeling and Sentiment analysis

Fei Liao*
School of Foreign Languages
Wuhan Business University
Wuhan, Hubei, China
20170141@wbu.edu.cn

Abstract

As one representative of autonomous vehicles (AVs), Apollo Go, Baidu's self-driving cab service, has been operated in 11 cities in China and aroused heated discussion on Chinese social media after its crash with one pedestrian in Wuhan in July, 2024. It's significant to explore the public's opinions and sentiment for a company to promote its products and expand its market. Chinese social media, Weibo, provides a vast amount of data to dig out the public attitudes and perspectives. In this paper, 11,882 Weibo posts from May 29th to August 15th 2024 were crawled for topic modelling and sentiment analysis by using LDA algorithm and sentiment dictionaries. It is found that the public mainly focus on five topics related with Apollo Go, including unemployment, accident and privacy, rumor response and market expansion, industry development and riding experience as well as price. While some people embrace Apollo Go which, they think, stands for the future means of transportation, some of them show their concern about the liability for the accident, riding safety, possible privacy invasion as well as the potential unemployment. At the end this study, several measures are put forward to tackle those negative sentiments to promote the popularity of AVs. Meanwhile, the limitation of this study were also proposed at the conclusion of this paper.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing; Computing methodologies; Machine learning; Machine learning algorithms.

Keywords

Autonomous vehicles, Apollo Go, Topic modelling, Sentiment analysis, Weibo

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*Corresponding author.

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1 Introduction

With the technological advancements such as artificial intelligence, deep learning, automatic braking systems, cruise control and so on, autonomous vehicles (AVs), defined as vehicles capable of sensing their environment and operating without human involvement [1], have gradually entered the market. Many companies, like Tesla, Baidu, Waymo, Cruise, are competing for the consumers. Therefore, it's crucial and meaningful for them to collect the public's perceptions and feedback about AVs, which can help developers to improve and promote the product. What's more, it will inform the policy makers to implement wise rules and strategies.

As a pioneer of AVs, Apollo Go, Baidu's self-driving cab service, marks a significant step towards the commercialization of autonomous driving technology in China. Based on Baidu Apollo's Level 4 autonomous driving technology, Apollo Go is capable of handling complex urban road scenarios, including high-precision perception, intelligent decision-making, and precise control. As of July 2024, Apollo Go has opened manned testing and operation services in 11 cities, with cumulative orders exceeding 6 million. However, while gaining the recognition from certain consumers, Apollo Go has sparked some controversies because of its safety concern, low-price policy and the potential impact on the livelihood of traditional taxi and ride-hailing drivers. In July 2024, Apollo Go made several appearances on the Weibo trending list. It's absolutely significant and necessary for Apollo Go to gain the public's opinions and perceptions about its product especially on this development stage.

Compared with traditional survey tools such as polls and telephone interviews [2], social media platforms, like Twitter, Facebook, Weibo, are more cost-effective, time-saving and convenient with its enormous data, true expression and real-time content. Launched in August 2009 by Sina Corporation, Weibo has grown to become one of the most popular social media platforms in China, with millions of active users, who can post short messages, imagines, and videos to share their opinions and engage with other users through likes, reposts and comments. Therefore, Weibo has offered us an abundance of comprehensive data to conduct opinion mining.

In recent years, some researchers have studied the public perceptions of AVs, including the sentiments towards AVs, the willingness to use AVs, the factors influencing the adoption and so on. Most of them use traditional survey tools, such as questionnaires and interviews. For example, Hudson, Orviska and Hunady [3] survey 1000 people in each EU country to uncover the determinants of

individual attitudes to AVs, both cars and trucks; Wang et al. [4] survey 834 US-based participates to examine how attitudinal factors shape AV acceptance across a range of AV "modes". Hilgarter and Granig [5]conduct an in-depth, semi-structured interviews with 19 participants aged between 20 and 75 years in Austria to reveal in-depth insights into AV integration, exploring public perceptions of AVs. Besides, some researchers also adopt social media platform to explore the public perception of AVs. For instance, Io, Lee and Lian [6] has crawled 10,374 AV-related comments from "hot posts" in Weibo and used sentiment analysis to evaluate people's feelings and perceptions of AVs. Penmetsa et al. [1] conduct before-and-after sentiment analysis to examine how the two fatal crashes, namely, an Uber-pedestrian crash and a Tesla's Model X crash have affected people's perceptions of self-driving and autonomous vehicle technology using Twitter data. Dutta et al. [7] examine 25,000 comments of 15 most-viewed videos about AVs on YouTube to reveal how people viewed AVs and what polarities exist regarding content and automation level.

However, so far, there are few studies talking about different sentiments for different topics related with AVs. Moreover, there are no studies which focus on the public's perception of Apollo Go.

Therefore, in view of the leading role of Apollo Go in AVs industry as well as the rich data and significant influence of Chinese Weibo, this study, by using Weibo's posts, aims to uncover the hidden topics discussed, explore sentiments towards Apollo Go in terms of different topics, and discuss sentiment variations resulted from different social events.

The major contributions of the paper are as follows:

On the one hand, by examining sentiment variations towards Apollo Go based on Weibo data, this paper compensates for the current studies on AVs. On the other hand, with hidden social events analysis, preferences and concerns exploration as well as corresponding suggestions, this study also provides valuable insight into the current promotion and deployment of AVs.

2 Materials and Methods

This study mainly employed topic modeling and sentiment analysis to delve into the public's perspectives and feelings towards Apollo Go.

2.1 Data collection and Pre-processing

With the keyword "萝卜""萝卜快跑"(Apollo Go), we used Octopus data scraping tool to retrieve Weibo posts from May 29, 2024 to August 15, 2024. The reason why we chose May 29 is because on this day Apollo Go officially responded to the rumors on his Weibo official account. At the same time, screen name and posted time were also scraped. Initially, 50,977 posts were retrieved. However, some posts containing "萝卜" were irrelevant with the autonomous diving Apollo Go. Instead, they might talk about other things like the vegetable radish, the tool Radish knife ("萝卜刀"). So we further extracted the posts contained with key phrase "萝卜快跑"(Apollo Go) or "无人驾驶"(autonomous driving) or "无人车"(autonomous car) or "智能驾驶"(intelligent driving) or "苕"(stupid) . "苕" belongs to Wuhan dialect. Some Wuhan netizen banteringly called Apollo Go as "苕萝卜"(stupid Apollo Go) to describe its oversensitiveness, slowness or mistakes made on the way.

To ensure the data quality,we removed the posts related with advertisements and stock which were obviously irrelevant with the common public as well as negatively influenced our topic and sentiment analysis. Next, we deleted the duplicate posts, Emojis, @ user name, and URLs. That is, we only retained the literal texts.

Subsequently, we adopted the Jieba Python toolkit for word segmentation. To eliminate words that were commonly used yet didn't aid in differentiating one post from another, we employed a set of stop words developed by the Harbin Institute of Technology. Additionally, we specifically removed "萝卜" and "萝卜快跑" which were prevalent in our dataset. Last, only content words like nouns, verbs and adjectives were retained as feature words.

Ultimately, we have 11,882 Weibo posts for our following LDA topic modeling and sentiment analysis.

2.2 Data analysis

2.2.1 Topic Modeling with Latent Dirichlet Allocation. Topic modeling, one of Natural Language Processing (NLP) techniques, is used to discover the hidden common topics in a large corpus. LDA (Latent Dirichlet Allocation), one unsupervised learning algorithm, is the most commonly and effectively used topic modeling technique to detect the topics from social media [8]. It assumes that each document contains a certain number of topics, while each topic is represented by a set of words [9].

In this study, the "lda" package in Python was employed for topic modeling and the LDAvis package for LDA visualization. Determining the number of topics is critical for the model performance. We combined the perplexity curve and the bubble plot to figure out the optimal number. Generally speaking, the smaller the perplexity value is, the stronger the generative capacity of the topic model is. That is, when the perplexity value is minimized, the topic has a better semantic expression effect. So usually we choose the number when the perplexity value is the smallest. What's more, no or few overlaps of the bubble plot indicates high topic distinctiveness with good model performance.The equation 1) is used to calculate the perplexity value.

$$perplexity(D) = \exp \frac{\sum_{d=1}^{M} \log P(W_d)}{\sum_{d=1}^{M} N_d} \tag{1}$$

In the equation 1), D represents the collection of all words in the document; M is for the number of documents; W_d represents the word in document d; N_d means the number of words in each document d; and $P(W_d)$ is the probability of a word appearing in the document.

Firstly, we adopted the comprehensive approach to tune and train our model. The hyperparameters were adjusted to get the optimal number of topic. In our study, when alpha was 1.2 and beta was 0.3, we got relatively clear topic partitions in the bubble pot where the optimal number was 5 which was also the point when the perplexity value was minimized at 727.53.

Next, we exported topic distribution for each document and the word distribution for each topic. Then we selected top 30 words for each topic.

Finally, by reading those representative words and corresponding posts, we identified and labeled each topic to get the public's perspectives and focuses on Apollo Go.

2.2.2 Sentiment analysis with self-built and other widely recognized sentiment dictionaries. Sentiment analysis, which is also known as opinion mining, is a subfield of NLP. With sentiment analysis, we can unearth the users' opinions, emotions, attitudes towards the product to aid the market expansion. Traditionally, there are three ways to conduct sentiment analysis, namely, machine learning approach ; lexicon-based analysis; and deep learning approach.

In this study, we utilized a dictionary-based approach to uncover the public's hidden sentiments towards Apollo GO. The reason why we chose this approach was that it entailed no manual annotation, which means it was highly efficient and cost-effective. The construction of sentiment dictionary and the calculation process were as follows:

Firstly, we constructed our sentiment dictionary with one list of positive emotion words and the other of negative emotion words. These emotion words were primarily from four widely used emotion dictionaries, namely, Hownet of Chinese CNKI, NTUSD of Taiwan University, Emotion dictionary of Li Jun of Tsinghua University and the Chinese DLUT-Emotion Ontology. However, given that new cyber language and some Wuhan dialect also expressed the users' emotions but were not included in the above four dictionaries, we expanded our dictionary by adding another 181 positive emotion words and 52 negative emotion words, which were mainly from emerging internet language and local dialect such as "草台班子"(fly-by-night operation), "爆火"(go viral), "牛"(awesome), "地表最强"(the most powerful), "碰瓷"(blackmail),"苕"(stupid) and so on. By doing this, we further improved the accuracy of sentiment recognition.

Next, we created two lists of negation words and adverbs of degree, because the negation words like "不" (not), "没有"(no) can reverse the sentiment polarity while the adverbs of degree will impact intensity of sentiment. According to the intensity, the adverbs of degree were classified into four levels, titled as"extreme", such as "最"(most), "绝对"(absolutely); "very", such as "非常"(very), "贼"(really); "more", such as" 格外"(particularly), "不过"(just); and "ish" such as "略微" (slightly), "有点" (a little).

Finally, we calculated the sentiment score for the post. At the very beginning, we need to assign certain points and weights for emotion words, negation words and adverb of degree. Each emotion word, both negative and positive, in the dictionary was assigned 1 point, while each negation word was weighted as -1. What's more, different weights were allocated to four levels of adverbs of degree, "extreme" for 4 , "very" for 3, "more" for 2 and"ish" for 0.5. Then, we employed Python to read all the emotion words, negation words as well as the adverbs of degree for calculation. The equation 2) was adopted to calculate the total sentiment score for each emotion lexicon.

$$\mathrm{E}(w) = d(w)a(w)e(w) \tag{2}$$

In equation 2), d(w) represents the weight for negation words; a(w) means the accumulation of weight for all the adverbs of degree; e(w) is for the initial score for the emotion word; and E(w) represents the total score for each emotion lexicon.

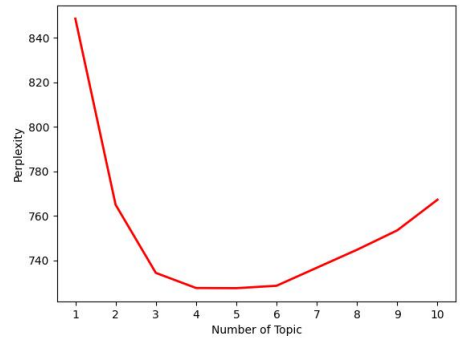

Figure 1: Perplexity-topic number line chart.

$$\mathrm{d}(w) = (-1)^n \tag{3}$$

$$a(w) = \sum_{i=1}^{t} ag_i \tag{4}$$

$$E(r) = \sum_{w \in r} \mathrm{E}(w) \tag{5}$$

In equation 3), n refers to the number of negation words preceding the emotion word. In equation 4), t denotes the quantity of adverbs of degree surrounding the emotion word, and ag_i signifies the weight assigned to the adverb of degree i. For each post contained several emotion lexicons, we determined the aggregate sentiment scores for both positive and negative sentiments in one post with equation 5). E(r) is for the total sentiment score of the positive or negative of one post while r stands for the set of positive or negative lexicons in one post.

Ultimately, we computed the sentiment scores for all positive and negative lexicons within a single post. The ultimate sentiment score for each post was then deduced based on equation 6).

$$\mathrm{E} = \frac{E(r)_+ - E(r)_-}{E(r)_+ + E(r)_-} \tag{6}$$

In equation 6), E(r)+ is the total score for all the positive lexicons in one post, and E(r)- for all the negative. E means the final sentiment score for one post. After calculation, the E value was ranged from -1 to 1, then we carried out data normalization from [-1,1] to [0,1] with MinMaxScaler algorithm. Then, we categorized posts with a score less than 0.5 as negative, those with a score of exactly 0.5 as neutral, and those with a score greater than 0.5 as positive.

Based on sentiment polarity of each post, we then analyzed the sentiment for each topic as well as the sentiment variations with critical social events.

3 Results and Discussion

3.1 Apollo Go Topics

According to the method described in Section II, combing the perplexity value as showed in Figure 1 and LDAvis bubble plot, we identified five latent topics.

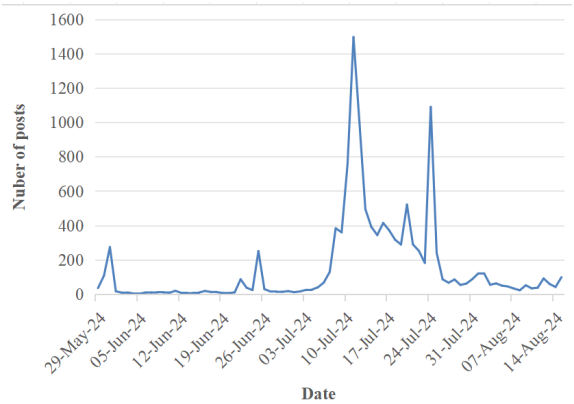

Figure 2: Temporal distribution of Apollo Go related posts.

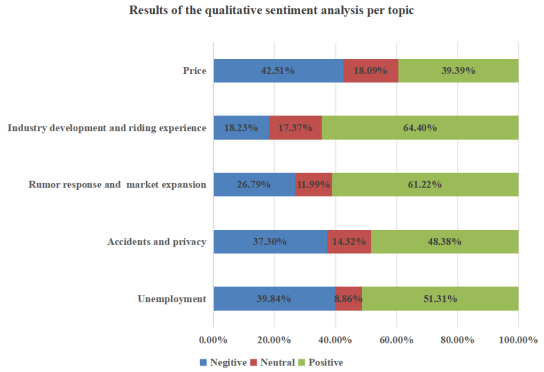

Figure 3: Results of the qualitative sentiment analysis per topic.

3.1.1 Unemployment. 31.9% of posts (n=3,793) talked about the potential unemployment which may be brought about by the rising of Apollo Go. We can derive this topic from some representative words like “饭碗”(job), “失业”(job loss) , “取代”(replace) and “出租车”(taxi). Some people expressed the Robotaxi may replace the traditional taxi and online taxi-hailing service, which would result in unemployment.

3.1.2 Accident and privacy. 18.6% of posts (n=2,220) expressed views about Apollo Go’s crash with a pedestrian in July 2024 and the camera installment in the car. We summarized the topic from such representative words as “事故” (accident), “行人”(pedestrian), “闯红灯”(run the red right) and “摄像头”(camera) . Weibo users talked about the liability of the accident while some shared their opinions on the topic if the camera invaded the riders’ privacy.

3.1.3 Rumor response and market expansion. There were about 14.3% of posts(n=1,702) discussing the rumor faced by Apollo Go and its response as well the as its market expansion to other cities. We got the topic from the key words like “恶意”(malicious), “抹黑”(defame), “谣言”(rumor), and “城市”(city), “上海”(Shanghai).

3.1.4 Industry development and riding experience. About 25.6% (n=3045) of users discussed the Robotaxi’s industrial development and shared their riding experience. They talked the Apollo Go’s competitors such as Tesla, the autonomous technology while some shared personal riding experience.

3.1.5 Price. 9.4% of post(n=1,122) were about the price of Apollo Go. People showed attitudes towards the low price of Apollo Go at the very beginning as well as its rise of price. The topic was inferred from the representative words like “涨价”(rise of price), “扰乱”(disrupt), “优惠”(discount).

By analyzing the temporal distribution of the related posts as shown in Figure 2, we can find that the first peak appeared on May 31st. At the end of May, some netizen complained on Wuhan City Message Board that Apollo Go moved slowly within the city area and parked without cause, leading to prolonged and extensive traffic congestion for subsequent vehicles. On May 29th, The official Weibo Account of Apollo Go replied that such complaints were not square with facts. The public shared their opinions on the complaints and the official response. The second peak on June 25th , also resulted from the Apollo Go’s rumor response to a case that suspects from more than 10 MCN organizations were suspected of manipulating hundreds of accounts to post slanderous remarks about Apollo Go. Most Weibo users showed their anger toward the suspects. The Robotaxi’s crash with a pedestrian on July 7th gave rise to the highest peak on July 10th . People expressed their concern about the safety of autonomous driving and the accident liability. Moreover, many people discussed the potential unemployment caused by the market expansion of Apollo Go, who claimed it would make profit next year and would expand its market to other cities in China. Besides, the rumor raged that Apollo Go had exceeded a size of 1,000 vehicles in Wuhan. In the next week, several hashtags about Apollo Go appeared. People talked about the low price of the Robotaxi as well as its order explosion. On July 24, Apollo Go gained the second highest peak for its price rising. People talked about the reasons behind the rising while expressing their anger or understanding. At the same time, people compared Apollo Go with Tesla and showed their preference as well as discussed the autonomous driving technology.

3.2 Sentiment analysis

In the following part, the sentiment and attitudes for each topic will be analyzed.

Figure 3 indicates the sentiment polarity for each topic. It can be seen that price got the highest percentage of negative sentiment, which accounted for 42.51%. According to the posts, some users thought the undercutting prices disrupted market order, and monopolized the market, which would undermine such competitors as traditional taxis and onlineride-hailing. What’s more, at the end of July, when Apollo Go raised its price, people’s negative sentiments become stronger. They criticized its act that they entered the market with low prices, and then raised the prices to make a profit. However, we can still see 39.39% posters expressed their support for Apollo Go’s price. On the one hand, they visioned the bright future of their using AVs to facilitate their transportation and make money while enjoyed the promotional price of Apollo Go. On the other hand, they thought that the price was determined by the market, and raising price was just the normal operation of the

enterprise, who needed to make profit to survive in the competitive market.

The second largest negative sentiment proportion, 39.84%, stems from the unemployment. Many people worried that Apollo Go might taken away job opportunities from ordinary people and some people even expressed their anger toward such technology. By contrast, about half of the posts showed positive attitudes. They believed that autonomous driving was an inevitable trend, and at the same time, they compared autonomous driving to the first and second industrial revolutions, thinking that new job positions would emerge and we should view it rationally.

As for the accidents and privacy, 37.30% posters expressed their anger towards the behavior of pedestrians running red lights as well as challenged the liabilities of AVs. They proposed the question that who should be responsible for the accident in such autonomous vehicle accident. What's more, some of them were afraid of the leakage of private data. However, some users thought Apollo Go abide by traffic rules, while expressing understanding towards the camera installment, which, in their opinion, could ensure the safety. Therefore, we can still see about 60% posters displayed positive or neutral sentiments in this topic.

The highest percentage of positive attitudes, 64.40%, came from the topic industry and riding experience. Many people embraced new technology and they believed AVs would became better in the future, finally becoming the mainstream transportation means. They thought nobody could stop this technological trend. In the meantime, some shared their riding experience, feeling it was convenient, cheap and safe, never refusing to carry passengers. However, some people criticized that Apollo Go was silly and unreliable, for it was slow and over sensitive to obstacles, which had resulted in the traffic congestion. What's worse, a few of them even expressed their safety worry, that was, they dared not to take Apollo Go.

As far as rumor response and market expansion is concerned, only 26.79% posts showed negative sentiments. While some displayed anger towards those rumors, like Apollo Go would carry out franchise recruitment and it led to serious traffic jam and other rumors, which attempted to bring shame on Apollo Go, some people said Apollo Go's expansion was disorderly expansion of capital, which would constitute unfair competition with unethical risk. Yet, there were 61.22% posts showing their support for Apollo Go's market expansion and hoped that we could be more tolerant of such new technology. What's more, some people gave the police a thumbs up for clearing the air for Apollo Go and called up everyone to resist the rumors.

4 Conclusions

In conclusion, the public mainly focus on five Apollo Go related topics, namely, unemployment, accidents and market expansion, rumor response and market expansion, industry development and riding experience, and price. Generally speaking, most people embrace this new technology for its convenience. However, some people hold opposing view. They worry the job loss, accident liability as well as the possible monopolistic market. This study has offered insights for the future deployment and promotion of AVs.

Firstly, as for the public's concern about safety and privacy, the developers should have thoroughly tests for AVs before they run on the road. Also, they need to inform the public of the safety standard to gain their trust. Secondly, with regard to job loss concern, businesses and governments should work together to create more job opportunities related with AVs and provide training programs for workers to relieve their work anxiety. Third, the government should update the law to clear the legal responsibilities and liabilities when a AV crush occurs. Last, for those people who dare not take AVs, the company may publicize some practical instructions to ease their uncertainties, telling them how to take AVs and how to react when facing emergencies.

However, we have to admit that there are still some limitations of this study. For example, though we have tried our best to add other emotion words as many as possible mainly from emerging internet language and local dialect, the dictionary-based approach still can't ensure the sentiment of every post is correctly identified, for it may ignore the context and the nuances of the sentiment expressions, which, indeed, is one shortcoming for both dictionary and machine learning approach. It's suggested that, in the future, researchers implement interdisciplinary research by drawing lessons from linguistics which to some extent considers the context of a certain text and conducts specific language analysis.

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